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TOWER FURNITURE FOR PET ANIMAL

Abstract

Provided is a tower furniture for pet animal that is easy for installation. The tower furniture includes: a height-adjustable top plate **50** on an upper end portion of the support member **25** at an uppermost part; a thread through fitting **61** provided at an upper end of the support member **25** at the uppermost part; an adjustment fitting **62** provided at a lower surface of the top plate **50;** an elongated thread **63** configured to be screwed into the adjustment fitting **62** and a lower end portion configured to be inserted into the thread through fitting **61;** a first nut **64** configured to be screwed onto the elongated thread **63** and to press against an adjustment fitting **62** side; and a second nut **66** configured to be screwed onto the elongated thread **63** and to press against an upper end side of the support member **25.**

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to tower furniture for a pet animal, and particularly relates to a ceiling strut structure such as a cat tower.

BACKGROUND

[0002] In the related art, a cat tower that allows a domestic cat, who tends to lack exercise, to climb and play indoors. The cat tower includes a support formed by vertically connecting a plurality of support members with a table or the like sandwiched, and a base and a top plate provided at a lower end and an upper end of the support, and can be installed vertically between a floor and a ceiling. The top plate is provided in a height-adjustable manner with respect to the support member at an uppermost part. In a case of installing the cat tower, after assembling the cat tower, the cat tower is erected vertically on the floor with the height of the top plate slightly lowered, then the height of the top plate is adjusted, and the top plate is pressed against the ceiling and locked.

[0003] A height adjustment structure (ceiling lock structure) for a top plate **1** of the cat tower in the related art will be described with reference to FIGS. **1** to **3**. FIG. **1** is a partially omitted cross-sectional view showing an upper portion of the cat tower installed between the floor and a ceiling **15**, FIG. **2** is a cross-sectional view of the support member at the uppermost part in FIG. **1**, and FIG. **3** is an exploded view of a top plate height adjustment unit. In FIGS. **1** to **3**, a cylindrical lid member **3** made of plastic is fitted to an upper end portion of a support member **2** at an upper part, and a nut **4** as a female thread fitting is embedded in a lower surface side of a through hole **3***a* at a center of the lid member **3**. A disk-shaped upper end plate **5** made of wood is placed on the support member **2**. An unthreaded hole **5***a* is formed in a center of the upper end plate **5**, and the unthreaded hole **5***a*, the hole **3***a* of the lid member **3**, and the nut **4** are coaxially arranged.

[0004] The top plate **1** is made of wood, and an adjustment fitting **7** as a female thread fitting is attached to a center of a lower surface of the top plate **1** in a manner of protruding downward. The adjustment fitting **7** is a bottomed fitting with a female thread. An upper end portion of an elongated thread **8** is screwed into the adjustment fitting **7**. Two first nuts **10** are screwed onto an upper portion of the elongated thread **8** with a washer **9** sandwiched so as to press against the adjustment fitting **7**. A lower end portion of the elongated thread **8** is screwed into the nut **4** through the unthreaded hole **5***a* of the upper end plate **5** and the hole **3***a* of the lid member **3**. Two second nuts **12** are screwed onto a lower portion of the elongated thread **8** with a washer **11** sandwiched so as to press against the upper end plate **5** downward. A reaction force received from the upper end plate **5** is transmitted to the top plate **1** via the elongated thread **8**, and the top plate **1** is pressed against the ceiling **15** and locked.

[0005] When a method for adjusting a height of a top plate in the related art is described with reference to FIGS. **4** to **11** and FIG. **1**, first, the two first nuts **10** are screwed onto the upper portion of the elongated thread **8** with the washer **9** sandwiched, the two second nuts **12** are screwed onto the lower portion with the washer **11** sandwiched, and a lower side of the elongated thread **8** is inserted into the unthreaded hole **5***a* of the upper end plate **5** ((**1**) in FIG. **4**). Then, the upper end portion of the elongated thread **8** is screwed into the adjustment fitting **7** of the top plate **1**, the two first nuts **10** are turned with a wrench and are pressed strongly against the adjustment fitting **7**, and the top plate **1** and the top plate height adjustment unit are integrated ((**2**) and (**3**) in FIG. **4**).

[0006] Next, the lower end portion of the elongated thread **8** attached to the top plate **1** is inserted into the hole **3***a* of the lid member **3** of the support member **2** at the uppermost part of the cat tower (in this case, the upper end plate **5** goes down the elongated thread **8** and comes into contact with an upper end surface of the support member **2**), and while rotating the top plate **1** and the elongated thread **8** together about an axis, the lower portion of the elongated thread **8** is screwed into the nut **4** (FIGS. **5** and **6**). Accordingly, the top plate **1** is in a state of being supported by the support member **2**. The top plate **1** is rotated many times in a downward direction, and the nuts **12** come close to the upper end plate **5** (FIG. **7**).

[0007] Once the cat tower is erected vertically on the floor, if a distance between the top plate **1** and the upper end of the support member **2** at the uppermost part is too large and the top plate **1** tends to abut the ceiling **15**, the two second nuts **12** are turned to move to the upper portion of the elongated thread **8** (FIG. **8**), the top plate **1** is turned in the same direction as before to descend (FIG. **9**). On the other hand, if the distance between the top plate **1** and the upper end of the support member **2** is too small, the top plate **1** is turned in an opposite direction to rise.

[0008] In this way, once the top plate **1** is at an appropriate height where the top plate **1** does not abut the ceiling **15**, the cat tower is erected vertically on the floor, and the top plate **1** is turned in the opposite direction many times to raise the height of the top plate **1** by an operator standing on a stepladder (FIG. **10**). Then, when the top plate **1** comes into contact with the ceiling **15** (FIG. **11**), the second nuts **12** are turned with a wrench and tightened to the upper end plate **5**, and the first nuts **10** are also tightened to the adjustment fitting **7**. The reaction force received from the upper end plate **5** is transmitted to the top plate **1** via the elongated thread **8**, and the top plate **1** is pressed against the ceiling **15** and locked (FIG. **1**).

[0009] However, in the structure of the above-described top plate height adjustment unit in the related art, when the top plate **1** is supported by the support member **2** in advance, or when the height of the top plate **1** is adjusted by standing on a stepladder after the cat tower is erected on the floor, the top plate **1** with an elongated thread and the second nuts **12** have to be rotated many times, which is a troublesome and time-consuming operation.

SUMMARY

[0010] The present invention has been made in view of the above-described problems in the related art, and an object of the present invention is to provide tower furniture for a pet animal that does not require much effort for installation.

[0011] An invention according to claim **1** relates to indoor tower furniture for a pet animal, the tower furniture including: [0012] a support formed by vertically coupling a plurality of support members; [0013] a height-adjustable top plate supported by the support member at an uppermost part; [0014] an unthreaded hole portion provided at an upper end portion of the support member at the uppermost part; [0015] a female thread fitting provided at a lower surface of the top plate; [0016] an elongated thread having an upper end portion configured to be screwed into the female thread fitting and a lower end portion configured to be inserted into the unthreaded hole portion; [0017] a first nut configured to be screwed onto an upper portion of the elongated thread and to press against a female thread fitting side or a lower surface side of the top plate; and [0018] a second nut configured to be screwed onto a lower portion of the elongated thread and to press against an upper end side of the support member at the uppermost part.

[0019] In an invention according to claim **2**, [0020] an upper end plate is placed on an upper side of the support member at the uppermost part, [0021] the unthreaded hole portion configured to allow the lower portion of the elongated thread to be inserted is provided in the upper end plate, and [0022] the second nut screwed onto the elongated thread on an upper side of the upper end plate is configured to press against the upper end side of the support member at the uppermost part via the upper end plate.

[0023] In an invention according to claim **3**, [0024] one or more grooves are provided in a side circumferential surface of the second nut to allow rotation about an axis by inserting a flathead screwdriver.

[0025] In an invention according to claim **4**, [0026] a base is attached to a lower side of the support member at a lowest part, and [0027] an earthquake-proof gel member is provided on each of a lower surface of the base and an upper surface of the top plate.

[0028] According to the present invention, when installing the tower furniture for a pet animal between a floor and a ceiling, for example, the upper end portion of the elongated thread whose upper portion is screwed by the first nut and whose lower portion is screwed by the second nut is screwed in advance into the female thread fitting on the lower surface of the top plate to integrate the top plate and the elongated thread, and the lower portion of the elongated thread with the top plate attached is straight inserted, without turning, into the unthreaded hole portion provided in the upper end of the support member at the uppermost part in advance. After the tower furniture is erected vertically, an operator stands on a stepladder to pull up the top plate from the support member at the uppermost part straight along with the elongated thread without rotating the top plate about an axis. When the top plate comes into contact with the ceiling, the top plate can be easily locked to the ceiling by turning the second nut to press against the upper end side of the support member at the uppermost part, and thus it is unnecessary to turn the top plate and the elongated thread many times to adjust a height of the top plate, and an installation operation can be easily performed in a short time.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0029] FIG. **1** is a cross-sectional view in which a part of an upper portion of a cat tower in the related art is omitted;

[0030] FIG. **2** is a cross-sectional view of a support member in FIG. **1**;

[0031] FIG. **3** is an exploded view of a top plate height adjustment unit in FIG. **1**;

[0032] FIG. **4** is a diagram illustrating a method for assembling the top plate height adjustment unit in FIG. **1**;

[0033] FIG. **5** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **1**;

[0034] FIG. **6** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **1**;

[0035] FIG. **7** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **1**;

[0036] FIG. **8** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **1**;

[0037] FIG. **9** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **1**;

[0038] FIG. **10** is a diagram illustrating a method for adjusting a height of a top plate in FIG. **1**;

[0039] FIG. **11** is a diagram illustrating a method for adjusting the height of the top plate in FIG. **1**;

[0040] FIG. **12** is an external view of a cat tower according to an embodiment of the present invention;

[0041] FIG. **13** is a diagram illustrating a method for assembling a lower portion in FIG. **12**;

[0042] FIG. **14** is a diagram illustrating the method for assembling the lower portion in FIG. **12**;

[0043] FIG. **15** is a diagram illustrating the method for assembling the lower portion in FIG. **12**;

[0044] FIG. **16** is an exploded view of a top plate height adjustment unit in FIG. **12** and a cross-sectional view of a support member at an uppermost part;

[0045] FIG. **17** is a diagram illustrating a method for assembling the top plate height adjustment unit in FIG. **12**;

[0046] FIG. **18** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **12**;

[0047] FIG. **19** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **12**;

[0048] FIG. **20** is a diagram illustrating a method for adjusting a height of a top plate in FIG. **12**;

[0049] FIG. **21** is a diagram illustrating a method for adjusting the height of the top plate in FIG. **12**;

[0050] FIG. **22** is a diagram illustrating the method for adjusting the height of the top plate in FIG.

12;

[0051] FIG. **23** is a diagram illustrating the method for adjusting the height of the top plate in FIG. **12**;

[0052] FIG. **24** is a diagram illustrating the method for adjusting the height of the top plate in FIG. **12**;

[0053] FIG. **25** is a diagram illustrating the method for adjusting the height of the top plate in FIG. **12**;

[0054] FIG. **26** is a cross-sectional view of an upper portion of a cat tower according to a modification of FIG. **12**;

[0055] (**1**) in FIG. **27** is an exploded view of a top plate height adjustment unit in FIGS. **26**, and (**2**) and (**3**) in FIG. **27** are diagrams illustrating a method for assembling the top plate height adjustment unit;

[0056] FIG. **28** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **26**;

[0057] FIG. **29** is a diagram illustrating the method for assembling the top plate height adjustment unit in FIG. **26**;

[0058] FIG. **30** is a diagram illustrating a method for adjusting a height of a top plate in FIG. **26**;

[0059] FIG. **31** is a diagram illustrating the method for adjusting the height of the top plate in FIG. **26**; and

[0060] FIG. **32** is a diagram illustrating the method for adjusting the height of the top plate in FIG. **26**.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0061] Hereinafter, a best mode of the present invention will be described based on embodiments.

First Embodiment

[0062] Hereinafter, an embodiment of the present invention will be described with reference to FIGS. **12** to **25**.

[0063] FIG. **12** is an external view of a cat tower according to the embodiment of the present invention, FIGS. **13** to **15** are diagrams illustrating a method for assembling a lower portion of the cat tower, FIG. **16** is an exploded view of a top plate height adjustment unit in FIG. **12**, FIGS. **17** to **19** are diagrams illustrating a method for assembling an upper portion of the cat tower, and FIGS. **20** to **25** are diagrams illustrating a method for adjusting a height of a top plate.

[0064] In the present embodiment, an example in which an upper end plate **5** in FIG. **1** is omitted will be described.

[0065] In FIG. **12**, a reference numeral **20** represents a cat tower that is installed indoors to allow a cat to climb and play indoors. The cat tower **20** includes a support **30** formed by vertically and linearly connecting five cylindrical support members **21** to **25** which sandwich teardrop-shaped tables **26** to **28** and a spaceship hammock plate **29** with a hemispherical transparent capsule, a rectangular base **40** attached to a lower end of the support member **21** at a lowest part of the support **30**, a top plate **50** provided on an upper end side of the support member **25** at an uppermost part of the support **30**, and a top plate height adjustment unit **51** provided between the support member **25** and the top plate **50**.

[0066] A structure and an assembly method of a lower portion of the support **30** will be described with reference to FIGS. **13** to **15**. In each of the cylindrical support members **21**, **22**, **23**, **24**, low-height cylindrical lid members **31**, **32** made of plastic are respectively fitted to upper and lower end portions, and female thread fittings **33**, **34** implemented by nuts are respectively attached to back sides of holes **31***a*, **32***a* formed in centers of the lid members **31**, **32**. As shown in FIG. **16**, in the cylindrical support member **25** at the uppermost part, a high-height cylindrical lid member **60** made of plastic is fitted to an upper end portion, a low-height cylindrical lid member **34** made of plastic is fitted to a lower end portion, and a cylindrical and elongated thread through fitting **61** serving as an unthreaded hole portion is attached at a center of the lid member **60** on an upper side.

[0067] After a bolt **43** is inserted into a hole **41** in a center of the rectangular base **40** from below with a washer **42** sandwiched, the base **40** is fixed to the lower end of the support member **21** by screwing a male thread of the bolt **43** into the female thread fitting **34** (see FIGS. **13** and **14**).

[0068] After a lower portion of a short thread **80** is screwed into the female thread fitting **33** of the lid member **31** at the upper end of the support member **21**, a hole **26***a* formed near one end of the table **26** is passed through the short thread **80** from above. The female thread fitting **34** of the lid member **32** at the lower end of the support member **22** at a second part is screwed into an upper portion exposed above the short thread **80**, so that the support members **21** and **22** are linearly connected with the table **26** sandwiched (see FIGS. **14** and **15**). Similarly, on the support member **22**, the support members **23**, **24**, **25** are connected with the tables **27**, **28** and the spaceship hammock plate **29** sandwiched, so that the long linear support **30** is formed (see FIG. **12**).

[0069] The top plate **50** is made of wood, and an adjustment fitting **62** as a female thread fitting is attached to a center of a lower surface of the top plate **50** in a manner of protruding downward. The adjustment fitting **62** is a bottomed fitting with a female thread.

[0070] A height of the top plate **50** with respect to the support member **25** can be adjusted by the top plate height adjustment unit **51** provided between the support member **25** at the uppermost part and the top plate **50**. Specifically, as shown in FIG. **16**, the top plate height adjustment unit **51** includes an elongated thread **63**, two first nuts **64**, **64** provided at an upper portion of the elongated thread **63**, a spring washer **65**, and one second nut **66**, a spring washer **67**, and a washer **68** which have large diameters and are provided at a lower portion of the elongated thread **63**. In the second nut **66**, an inner side is made of a metal member, an outer side is a plastic member that can be grasped by a hand, and grooves **66***a* in a vertical direction are formed on a circumferential side surface at three locations spaced apart by 120 degrees in a circumferential direction. By inserting a flathead screwdriver into the groove **66***a*, the second nut **66** can be rotated about an axis.

[0071] The upper portion of the elongated thread **63** can be screwed into the adjustment fitting **62** of the top plate **50**, while the lower portion of the elongated thread **63** can be inserted into an unthreaded through hole **61***a* of the thread through fitting **61** attached to the lid member **60** fitted to the upper end portion of the support member **25** at the uppermost part. The thread through fitting **61** has no female thread that is screwed to the elongated thread **63**, and is freely and straight inserted and removed in an axial direction without turning the elongated thread **63**.

[0072] Next, a method for assembling an upper portion of the cat tower **20** and a method for adjusting the height of the top plate **50** will be described with reference to FIGS. **17** to **25**.

[0073] First, the two first nuts **64** are screwed onto the upper portion of the elongated thread **63** with the spring washer **65** sandwiched in advance, the second nut **66** is screwed onto the lower portion, the spring washer **67** and the washer **68** are fitted, and an upper end portion of the elongated thread **63** is screwed into the adjustment fitting **62** of the top plate **50**. Then, the first nuts **64**, **64** are turned with a spanner, and tightened and strongly pressed against an adjustment fitting **62** side, and the top plate **50** and the top plate height adjustment unit **51** are integrated (see FIGS. **17** and **18**).

[0074] Next, the lower end portion of the elongated thread **63** with the top plate **50** attached is inserted into the unthreaded through hole **61***a* of the thread through fitting **61** provided at the center of the lid member **60** at the upper end portion of the support member **25** at the uppermost part, which is placed horizontally on the floor. The second nut **66** comes into contact with the lid member **60** with the spring washer **67** and the washer **68** sandwiched. In this case, since the unthreaded through hole **61***a* does not have a female thread that is screwed to the elongated thread **63**, it is sufficient to simply insert the lower portion of the elongated thread **63** straight in the axial direction into the unthreaded through hole **61***a* of the thread through fitting **61** without turning the top plate **50** together with the elongated thread **63** about the axis many times (see FIGS. **18** and **19**).

[0075] If a distance between the top plate **50** and the upper end of the support member **25** is too large, and the top plate **50** is likely to collide with the ceiling **15** (see FIG. **12**) when the cat tower

20 is erected, the second nut **66** is turned to move to the upper portion of the elongated thread **63** (see FIG. **20**), and then the top plate **50** can descend together with the elongated thread **63** (see FIG. **21**).

[0076] In a state where the top plate **50** is at an appropriate height where the top plate **50** does not abut the ceiling **15**, the cat tower **20** is erected on a floor **16** (see FIG. **12**) with the base **40** put down, and an operator stands on a stepladder to pull up the top plate **50** together with the elongated thread **63** from the upper end portion of the support member **25** at the uppermost part, and to pull out the lower portion of the elongated thread **63** upward from the unthreaded through hole **61***a*. In this case, since the unthreaded through hole **61***a* does not have a female thread, it is sufficient to simply pull out the top plate **50** straight without turning the top plate **50** about the axis many times (see FIG. **22**). As the top plate **50** moves upward, the second nut **66** also moves upward.

[0077] When the top plate **50** comes into contact with the ceiling **15** (see FIG. **23**), the second nut **66** is turned by a hand to move to the lower portion of the elongated thread **63**, is tightened to the upper end side of the support member **25** at the uppermost part, and is pressed against the lid member **60** with the spring washer **67** and the washer **68** sandwiched. Accordingly, the top plate **50** presses against the ceiling **15** upward due to a reaction (see FIG. **24**).

[0078] Finally, a flathead screwdriver **70** is inserted into the vertical grooves **66***a* formed at three locations on the circumferential side surface of the nut **66** and is turned to further tighten the nut (see FIG. **25**), the top plate **50** is strongly pressed against the ceiling **15**, and the cat tower **20** is firmly fixed and installed between the floor **16** and the ceiling **15**. Accordingly, the top plate **50** can be easily locked to the ceiling **15**.

[0079] According to the present embodiment, the thread through fitting **61** without a female thread is provided in the center of the lid member **60** fitted to the upper end portion of the support member **25** at the uppermost part, the elongated thread **63** of the top plate height adjustment unit **51** can be inserted into and removed from the thread through fitting **61** straight in the axial direction without being turned about the axis, and when the cat tower **20** is installed between the floor **16** and the ceiling **15**, the upper end portion of the elongated thread **63** whose upper portion is screwed by the first nut **64** and whose lower portion is screwed by the second nut **66** is screwed in advance into the adjustment fitting **62** as a female thread fitting on the lower surface of the top plate **50**, the first nut **64** is tightened and the adjustment fitting **62** is pressed to integrate the top plate **50** and the elongated thread **63** in advance, and the lower portion of the elongated thread **63** with the top plate **50** attached is straight inserted, without turning, into the unthreaded through hole **61***a* of the thread through fitting **61** provided at the upper end of the support member **25** at the uppermost part in advance. After erecting the cat tower **20** vertically, the operator stands on the stepladder to pull up the top plate **50** straight together with the elongated thread **63** from the upper end portion of the support member **25** at the uppermost part without rotating the top plate **50** about the axis. When the top plate **50** comes into contact with the ceiling **15**, the second nut **66** is turned to move to the lower portion of the elongated thread **63** and is pressed against the upper end side of the support member **25** at the uppermost part, and if the top plate **50** is pressed against the ceiling **15** by a reaction, the top plate **50** is easily locked to the ceiling **15**, and thus it is unnecessary to turn the top plate **50** together with the elongated thread **63** many times to adjust the height of the top plate **50**, and the installation operation can be easily performed in a short time.

[0080] In the above-described embodiment, the elongated thread is inserted into the upper end of the support member at the uppermost part, but as in a cat tower **20**A shown in FIG. **26**, the disk-shaped upper end plate **5**, which is a separate member same as that shown in FIGS. **2** and **3**, may be placed on the support member **25** at the uppermost part, the lower portion of the elongated thread **63** may be removably inserted into an unthreaded hole (see the reference numeral **5***a* in FIG. **1**) as an unthreaded hole portion formed in the center of the upper end plate **5**, and when the second nut **66** is tightened, the upper end side of the support member **25** at the uppermost part may be pressed via the upper end plate **5**.

[0081] Specifically, as shown in (**1**) in FIG. **27**, the top plate **50**, the top plate height adjustment unit **51**, and the upper end plate **5** are prepared. First, the two first nuts **64** are screwed onto the upper portion of the elongated thread **63** with the spring washer **65** sandwiched, the second nut **66** is screwed onto the lower portion, the spring washer **67** and the washer **68** are fitted, and a lower side of the elongated thread **63** is inserted into the unthreaded hole **5***a* of the upper end plate **5** ((**2**) in FIG. **27**). Then, the upper end portion of the elongated thread **63** is screwed into the adjustment fitting **62** of the top plate **50**, the first nuts **64**, **64** are turned with a spanner, and tightened and strongly pressed against the adjustment fitting **62**, and the top plate **50** and the top plate height adjustment unit **51** are integrated (see (**3**) in FIG. **27**).
[0082] Next, the lower end portion of the elongated thread **63** with the top plate **50** attached is inserted into the unthreaded through hole **61***a* of the thread through fitting **61** provided at the center of the lid member **60** at the upper end portion of the support member **25** at the uppermost part, which is placed horizontally on the floor. The second nut **66** comes into contact with the lid member **60** with the upper end plate **5**, the spring washer **67**, and the washer **68** sandwiched. In this case, since the unthreaded through hole **61***a* does not have a female thread that is screwed to the elongated thread **63**, it is sufficient to simply insert the lower portion of the elongated thread **63** straight in the axial direction into the unthreaded through hole **61***a* of the thread through fitting **61** without turning the top plate **50** together with the elongated thread **63** about the axis many times (see FIGS. **28** and **29**).
[0083] In a state where the top plate **50** is at an appropriate height where the top plate **50** does not abut the ceiling **15**, the cat tower **20**A is erected on the floor **16** (see FIG. **12**) with the base **40** put down, and the operator stands on a stepladder to pull up the top plate **50** together with the elongated thread **63** from the upper end portion of the support member **25** at the uppermost part, and to pull out the lower portion of the elongated thread **63** upward from the unthreaded through hole **61***a*. In this case, since the unthreaded through hole **61***a* does not have a female thread, it is sufficient to simply pull out the top plate **50** straight without turning the top plate **50** about the axis many times. As the top plate **50** moves upward, the second nut **66** also moves upward (see FIG. **30**).
[0084] When the top plate **50** comes into contact with the ceiling **15** (see FIG. **31**), the second nut **66** is turned by a hand to move to the lower portion of the elongated thread **63**, is tightened to the upper end plate **5**, and is pressed against the lid member **60** on the upper end side of the support member **25** at the uppermost part via the spring washer **67**, the washer **68**, and the upper end plate **5**. Accordingly, the top plate **50** presses against the ceiling **15** upward due to a reaction (see FIG. **32**).
[0085] Finally, the flathead screwdriver **70** is inserted into the vertical grooves **66***a* formed at three locations on the circumferential side surface of the nut **66** and is turned to further tighten the nut (see FIG. **32**), the top plate **50** is strongly pressed against the ceiling **15**, and the cat tower **20**A is firmly fixed and installed between the floor **16** and the ceiling **15**. Accordingly, the top plate **50** can be easily locked to the ceiling **15** (see FIG. **26**).
[0086] An earthquake-proof gel member may be attached to each of an upper surface of the top plate **50** and a lower surface of the base **40**.
[0087] When the adjustment fitting **62** is embedded in a lower surface side of the top plate **50** and the first nut **64** screwed onto the elongated thread **63** is tightened, the first nut **64** may press against the lower surface side of the top plate **50**.
[0088] The present invention may also be used as tower furniture for a pet animal for other small animals such as puppies and hamsters other than cats.
[0089] The present invention can be applied to tower furniture for a pet animal on which cats, puppies, and the like kept indoors can exercise. [0090] **5** upper end plate [0091] **15** ceiling [0092] **16** floor [0093] **20**, **20**A cat tower [0094] **25** support member [0095] **30** support [0096] **50** top plate [0097] **51** top plate height adjustment unit [0098] **60** lid member [0099] **61** thread through fitting

[0100] **62** adjustment fitting [0101] **63** elongated thread [0102] **61***a* unthreaded through hole [0103] **66** second nut [0104] **66***a* groove

Claims

1. Indoor tower furniture for a pet animal, the tower furniture comprising: a support formed by vertically coupling a plurality of support members; a height-adjustable top plate on an upper end portion of the support member at an uppermost part; an unthreaded hole portion provided at the upper end portion of the support member at the uppermost part; a female thread fitting provided at a lower surface of the top plate; an elongated thread having an upper end portion configured to be screwed into the female thread fitting and a lower portion configured to be inserted into the unthreaded hole portion; a first nut configured to be screwed onto an upper portion of the elongated thread and to press against a female thread fitting side or a lower surface side of the top plate; and a second nut configured to be screwed onto a lower portion of the elongated thread and to press against an upper end side of the support member at the uppermost part, wherein the unthreaded hole portion has no female thread that is screwed to the elongated thread, and is freely and straight inserted and removed in an axial direction without turning the elongated thread.
2. The tower furniture for a pet animal according to claim 1, wherein an upper end plate is placed on an upper side of the support member at the uppermost part, the upper end plate is provided with the unthreaded hole portion configured to allow the lower portion of the elongated thread to be inserted, and the second nut screwed onto the elongated thread on an upper side of the upper end plate is configured to press against the upper end side of the support member at the uppermost part via the upper end plate.
3. The tower furniture for a pet animal according to claim 1, wherein one or more grooves are provided in a circumferential side surface of the second nut to allow rotation about an axis by inserting a flathead screwdriver.
4. The tower furniture for a pet animal according to claim 2, wherein one or more grooves are provided in a circumferential side surface of the second nut to allow rotation about an axis by inserting a flathead screwdriver.
5. The tower furniture for a pet animal according to claim 1, wherein a base is attached to a lower side of the support member at a lowest part, and an earthquake-proof gel member is provided on each of a lower surface of the base and an upper surface of the top plate.
6. The tower furniture for a pet animal according to claim 2, wherein a base is attached to a lower side of the support member at a lowest part, and an earthquake-proof gel member is provided on each of a lower surface of the base and an upper surface of the top plate.